

What the Endpoint Removes Before a Score Exists:

A Pre-Registered Study of Agentic Evaluation Across Four Models, Two Clouds and Two Tool Transports

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Abstract

A serving endpoint can remove a model from an evaluation before it produces a score, or destroy its measurements from inside one, and we document eight mechanisms with the endpoint's message for each. They are not equally easy to find: four are returned to an ordinary single-request capability check, three require a deliberately constructed conversational state, and one appears only when the model itself emits the output that triggers it. Ordinary capability checks are therefore not sufficient, and a harness certified healthy before collection can still lose a cell after the money is spent. Counted as model-platform pairs, eight of the 21 in the roster we assembled never produced a score — five of the 18 on the two platforms the design measures. The roster was assembled rather than sampled, so these are existence proofs with a stated denominator and not an incidence rate. **The transport moves a number, which is a bias. The endpoint deletes a row, which is a selection** — though not uniformly: some removals are access and lifecycle policies no pre-flight reaches, while others are collisions with a request contract the vendor documents, which a reader of the API reference would have caught before collecting.

That distinction is what the paper measures: four models, two managed clouds, two transports, 45 held-out reverse-engineering tasks, 5,809 trials. The principal hypotheses and the analysis plan were hash-frozen before collection; two of the ten tests were implemented under a timestamped amendment after collection had begun.

Second, temperature zero is not determinism: across the corpus, one model returns identical runs on 91–100% of binaries and another on 20–24%. That output distributions can differ between services at fixed model identity is already established; what §6 adds is one case where the difference reaches the *stability of a downstream agentic score*.

Third, the confirmatory family resolves no effect: no test passes its Holm threshold. The observed standard deviations span a factor of 23.7, so each contrast is reported with the precision implied by its own observed dispersion — what a design can separate is a property of the cell, not of the study.

What transfers is the checklist of Table 10 and the variance-decomposition method, not the coefficients. On the two contrasts with the largest between-task dispersion, the lever that binds is the number of tasks and not the number of runs, which inverts the calibration this design was sized on — so on a task family with this much between-task heterogeneity, a borrowed calibration sends a budget to the axis that does not bind.

Keywords large language models; tool calling; serving infrastructure; selection effects; experimental validity; reproducibility

Clinical trial number: not applicable.

1 Introduction

An agentic evaluation reports a number for a model. That number is produced by an apparatus: a transport that carries tool calls between the model and the harness, an endpoint that serves the weights, a set of tools with a particular return format, and a decoding configuration. The model’s name appears in the table. The apparatus does not.

This paper measures how much of that number the apparatus determines rather than the model, and finds two answers of different kinds. The transport moves a number, which is a bias. The endpoint deletes a row, which is a selection. A bias can be bounded, and a reader who knows its size can correct for it. A selection cannot, because the model that was refused has no line in the table to correct.

1.1 The confound this study removes

A companion study (Author(s) withheld, 2026) measured a 10.4pp drop when one model moved from native function calling to a textual protocol. It is a companion manuscript under double-anonymous review, so a reader cannot obtain it. Both figures this paper draws from it recompute from a snapshot of the exact cells they need, released in the deposit with their per-turn logs. On a single infrastructure that number admits two readings no further collection on that infrastructure can separate — a property of the model, or of that provider’s shim — because model and infrastructure are perfectly confounded when there is only one endpoint.

Serving the same nominal model from two managed clouds tests something a single endpoint cannot: whether the measured outcome changes across deployed services at fixed nominal model identity. It does not, by itself, say which component of a service produced any difference it finds — the two stacks differ in several observable quantities at once (§6.7) — but a difference that survives the change of service is a difference the model’s name alone does not describe. That cell — same nominal model, two commercial endpoints, transport manipulated as a paired factor, on a multi-turn downstream task — is what this study contributes as a design, and to our knowledge it has not been run.

1.2 Contributions

The order below is the one in which we would have a reader weigh them, not the one in which they were planned. **The first is the primary contribution: it does not rest on a p -value, and it is checkable by re-issuing the probes we release.**

1. **Eight mechanisms by which a serving endpoint removes a model from an evaluation before it produces a score, or destroys its measurements from inside one**, each with the endpoint’s message (§5). They fall into four operational classes — lifecycle and access policy, capability and API-contract constraint, catalogue and quota inconsistency, integration and adapter incompatibility — and they sit at three levels of detection difficulty, which is the operational finding: four are returned to an ordinary single-request capability check, three need a deliberately constructed conversational state, and one appears only when the model itself emits the output that triggers it. For the three that a single-request check does not reach we give the constructed probe that does, and for the last we give the trajectories in which it fired.
2. **Temperature zero is not determinism**, measured across the corpus: one model returns identical runs on 91–100% of binaries and another on 20–24% (§6). That output distributions can differ between services at fixed model identity is already established (Gao et al., 2025); in one of eight comparisons here the difference reaches the *stability of a downstream agentic score*, which is the extension this study contributes on that axis.

3. **A pre-registered $4 \times 2 \times 2$ design** — four models, two named managed clouds, two transports, 45 held-out binaries, 8 runs each — with the analysis script frozen by hash *together with* the hypotheses before collection began; two of the ten tests were implemented under a timestamped amendment after it had (§3). No test passes its Holm threshold. We report per contrast the effect its own observed dispersion would have let it resolve, because those dispersions span a factor of 23.7 and no single band describes them.
4. **A variance calibration that inverts on its own data.** This design was sized expecting run-to-run noise to dominate; measured here, the lever that binds is the number of tasks — 0.5% noise and a floor of 668 binaries on the contrast with the largest observed dispersion (§6.2). Whether such calibrations transfer between apparatuses is a question this raises and does not settle, and the stronger reading is a proposition rather than a result (§7.5).
5. **An artifact** carrying every raw row including the invalidated batches, the per-turn session traces, the frozen documents with their hashes, and the amendment log (§9).

1.3 The boundary of the claim

Three boundaries fix what follows. The transport contrast measures that the choice moves the number, not which choice is right. The eight mechanisms are existence proofs, each with its own denominator rather than a rate, and the class is open by construction. And the effect sizes belong to the scope §8 bounds.

The transferable output of this study is the checklist of Table 10 and the variance-decomposition method, not the coefficients. Two of the four models are open-weight, so the task, the corpus and the harness can be re-run on infrastructure a reader controls — though not this design: every cell here was served by one of the two commercial clouds, and a self-hosted re-run would be a different apparatus, which is precisely the property under study. The propositions of §7.5 are hypotheses this study instantiates, not claims it establishes. The narrower claim made here is that an evaluation which does not report its transport is not reproducible from its method section, whatever the task.

2 Related Work

This section positions the work against its neighbours and states the delta at the width the evidence supports.

2.1 Measurement bias, and the harness as its current object

The shape of our finding is old in systems research and new only in the evaluation of agents: that a measurement depends on the stack under it is unremarkable in performance work, where declaring the configuration is a norm. What agentic evaluation lacks is that norm. The argument has been made for the harness as a whole (Zhang et al., 2026; Yao et al., 2026); this study is downstream of it and supplies the measurement it lacks — what the undeclared choices are worth, one axis at a time.

Four lines of work bound this one, and the artifact expands each with the figures we take from it. **Apparatus variance is established methodology elsewhere:** rigorous performance measurement has long required reporting the configuration and treating run-to-run variation as first class (Georges et al., 2007; Iosup et al., 2011), and measurement bias from unreported setup choices was demonstrated two decades ago (Mytkowicz et al., 2009) — what is missing here is not the method but its application. **Non-determinism at nominal temperature zero** has been measured for repeated sampling (Atil et al., 2024) and traced

to the batch size a server happens to be running (He and Thinking Machines Lab, 2025) — the class of server-side quantity our §6.7 finds differing between two services at fixed model identity. **The endpoint is already known to be a service rather than an artifact:** a measurement study of hosted open-weight APIs finds one model differing across providers in protocol support, error semantics, context capacity and throughput (Li et al., 2026), which is our T5–T8 axis arrived at independently. **And the transport axis is visible to the field, in no fixed direction:** a leaderboard ranks every model in a native and a prompted mode (Patil et al., 2025); on one model the prompted variant is *ahead* by 6pp (bfc, 2026; bfc); on an agentic tool-use benchmark a prompt-based protocol severely degrades a model of the same family as our T3/T4, attributed to a template mismatch with the model’s training (Lei et al., 2025); and a third transport, tools as typed code stubs rather than JSON, matches or exceeds native calling on 11 of 14 models (Patel et al., 2026). Three transports, and no direction that survives a change of task family — which is what makes the axis a parameter to declare rather than a correction to apply. **And the task family has its own literature:** reconstruction from stripped binaries is active (Tan et al., 2024; Jiang et al., 2025), semantic equivalence under LLM-driven transformation has been evaluated directly, on a different corpus and task (De Tomasi et al., 2025), test-based equivalence has known limits (Liu and Wang, 2020), and the reproducibility of published results built on commercial models has been assessed directly, with model deprecation as a leading cause of failure (Angermeier et al., 2026).

2.2 The transport: known to move the number, in no fixed direction

The nearest thing to a counterexample is the leaderboard above: it ranks every model in both a native and a prompted mode, so the axis is not invisible to the field. It reports the two modes side by side rather than crossing them with the serving endpoint, and on one model the prompted variant is *ahead* by 6pp (bfc, 2026; bfc) — which we treat as a constraint on our claim rather than as support. Designing on an effect size estimated from earlier data is known to miss its target power (Albers and Lakens, 2018); what this case adds is a direction that reverses rather than drifts.

The reporting obligation is already established and we do not argue for it: community guidelines for empirical work with language models require the harness to be described when it goes beyond bare API calls (Baltes et al., 2026). Those guidelines do not make all of the apparatus fields examined here equally explicit, and Table 10 says which. What this study supplies is the other half: how much the choices in question move a measured number.

2.3 The endpoint: a unit of measurement, and a unit of selection

That two services can serve one model identity and return different output distributions is established, and measured at scale. A two-sample test on the Maximum Mean Discrepancy, applied to commercial inference APIs for four Llama models, finds **11 of 31 endpoints serving distributions that differ from the reference weights** — naming Amazon Bedrock and Azure AI Studio among the access paths, at a median 77.4% power against a range of distortions (Gao et al., 2025). That work settles the premise this study starts from, and it does so on raw output distributions from single prompts: it crosses no transport, runs no multi-turn agentic task, and says nothing about whether the divergence reaches a downstream score. What §6 adds is one case where it reaches the *stability* of such a score.

Holding weights, decoding and hardware fixed while varying only the inference engine produces measurable differences in output, which establishes the mechanism this study extends: if an engine moves a number at fixed weights, a managed platform choosing engine, version and scaffolding without disclosing any of them can move it too, and a reader has no way to know by how much (Pape et al., 2026).

Removal also breaks reproduction downstream, and its scale has been measured on one public arena’s own roster: of 243 models, 205 had been silently deprecated against 47 marked as such, at rates of 87.8% for open-weight and 80% for proprietary models (Singh et al., 2025). That audit names the mechanism — “Chatbot Arena may be forced to deprecate a model when a provider no longer supports it via its API” — and counts the outcome after the fact, aggregated over a roster; this census measures the refusals themselves, verbatim and case by case, before a score exists.

2.4 What the census adds to the failure taxonomies

The nearest neighbour to our census is a taxonomy of failure modes in multi-provider serving infrastructure, classified by origin layer and detectability (Pandey and Singh, 2026). Two of its five layers already reach failures occurring before generation — network and transport covers “TCP timeouts, TLS handshake failures, DNS resolution issues”, governance and cost covers “rate-limit counters that leak and never recover” — so the distinction we draw is not about *when* the failure lands.

What the census adds is a different question, asked one step earlier: not how a request fails once a model is being served, but *whether this endpoint will serve this model identity at all*, under this request contract, from this account and this region. Several of our mechanisms are access and lifecycle decisions rather than defects, and those persist for the duration of a study whoever is judged responsible; others — the request-contract constraints and the adapter incompatibility — could reasonably be filed as bugs by an operator who owns them. **Nothing in our contribution depends on which filing a reader prefers:** what §5.4 establishes is that mechanisms of this class exist, where in the evaluation lifecycle they act, what they cost a measurement, and that they need different detection. The artifact carries the layer-by-layer comparison for a reader who wants to place each row.

3 Method

This section fixes four things the rest of the paper rests on: what a trial measures, what the pre-registration froze together with the hypotheses, how a tool call travels under each transport, and which failures do not count as a measurement.

3.1 The task

Each trial gives a model one stripped x86-64 binary, decompiled with a static analysis server, and asks it to reconstruct compilable C source. The candidate is compiled in a pinned container and executed against the unit tests of the original program, and the score is the fraction that pass. The suites — five tests per binary — are inherited with the corpus rather than sized by us, and what that granularity costs the variance decomposition is measured in §6.1.

Pass-rate is a *lower bound* on semantic equivalence and it is reported as such throughout: a candidate correct but differing in an untested behaviour scores below one, and one wrong in an untested behaviour scores above its true worth. It is used because it is executable and adversarially hard to game, not because it is tight.

3.2 Pre-registration

The hypotheses, the test family, the falsifiers, and *the analysis script* were frozen by content hash on 2026-08-13 at 22:00, and the first data row of any batch is timestamped 32 minutes later; the first row of the batch this paper retains is 22 hours later. Both precedences are verifiable from the committed history. Registered reports have been available in this field since 2020 (Ernst and Baldassarre, 2023); what we add is the second half of that freeze.

Two of the ten tests carry a different status. T9 (heterogeneity across models) and T10 (transport $\times$ infrastructure interaction) belong to the family fixed at the freeze, but the frozen script emitted a non-computable p for them; they were implemented eleven hours after the freeze, under a dated succession document written *before* the change. Collection had begun by then, so their implementation is covered by a timestamped amendment rather than by the freeze. What is checkable is the state of the data at that moment: 861 rows, in three cells of a *single* model. T9 compares the transport effect *between* models and T10 needs both transports on both infrastructures, so neither test was computable from what existed. Both entered Holm at the $m = 10$ fixed before any data existed, so the correction they face was not chosen after seeing them. What was inspected before writing that document was the row count and the schema validity of what existed; no per-model result was computed or read. Every subsequent change to a frozen file is documented before the change and checked by a guard, and the register is in the artifact.

Table 1: The three batches the text names, and where each ended. The 861 rows visible when T9 and T10 were implemented belong to the batch later invalidated for the algorithm-name leak, so they are not part of either collection this paper reports.

batch	rows	where it ended
algorithm-name leak	2,390	invalidated and retained; the 861 rows visible at the T9/T10 amendment are inside it
Bedrock final-turn (M7)	453	invalidated and retained; that cell re-collected against a corrected client
original collection	5,805	reported throughout beside the primary basis
isolated-workspace re-collection	5,809	the primary basis (§6)

3.3 The two transports

The manipulated variable is how a tool call travels between the model and the harness. Both transports offer the same four tools, the same schemas, and the same budget: twelve exploration turns plus one mandatory final submission turn. A trajectory ends when the model submits a candidate through the `submit_candidate` tool, or when the budget is exhausted and the final turn is taken.

Native. The request carries a `tools` field. The model emits a structured tool-call object, which the runtime parses; results return in the role the provider’s API defines for tool output.

Textual. The request carries *no* `tools` field. The system prompt defines a line protocol: the model ends its reply with a single line `TOOL_CALL:` followed by a JSON object naming the tool and its arguments. The harness parses that line and returns the result as an ordinary user message.

Two consequences bound what the contrast means, and both belong before any number. First, the native transport admits more than one call per turn where the text protocol admits one by construction.

The confound is structurally available to every model and binds only where the model uses it, which two of four do not at all. Second, *a turn that produces prose without a tool call is not a failure*: it is a turn without a call and is recorded as such, since treating it as a terminal answer would end the trajectory at turn one and score the protocol rather than the model.

Table 2: Batching under the native transport, per model. The text protocol is fixed at one call per turn by construction, so the confound is present only where this rate exceeds one.

model	calls/turn	turns > 1	contrast affected
haiku	1.438	38%	T3
sonnet	1.333	33%	T4
gpt-oss	1.000	0%	—
llama	1.000	0%	—

Table 3: The six questions, what answers each, and its evidentiary status. Confirmatory items map onto hypotheses frozen before collection; the two tests behind RQ3 belong to that frozen family but were implemented under a timestamped amendment after collection had begun (§3). RQ4 was pre-registered as an exploratory census. RQ5 and RQ6 arose while the apparatus ran and are exploratory wherever they appear.

	question	answered by
RQ1	By how much does the transport of a tool call move the measured score, at fixed model and infrastructure?	H1, tests T1–T4 (<i>confirmatory</i>)
RQ2	By how much does the serving endpoint move it, at fixed model and transport?	H2, tests T5–T8 (<i>confirmatory</i>)
RQ3	Is the transport’s effect constant across models, and do transport and endpoint interact?	H3 and H4, tests T9–T10 (<i>confirmatory; implemented under amendment</i>)
RQ4	By which mechanisms does an endpoint remove a model from an evaluation before it produces a score?	the census, §5 (<i>exploratory, pre-registered as such</i>)
RQ5	How much of the transport difference is the loss of call batching rather than the change of format?	the constrained-native arm, §6.5 (<i>exploratory, arose during execution</i>)
RQ6	Does a variance calibration transfer between apparatuses that share model, corpus and metric?	§6.2 (<i>exploratory, arose during execution</i>)

3.4 What counts as a measurement

One predicate decides whether a row is a measurement, and collector and analysis use the same one, because two definitions of validity, one per half of a pipeline, diverge silently. It has two clauses, and they are stated here because every number in this paper is computed over the rows they admit: a row is a measurement unless the endpoint reported an infrastructure failure, or the harness itself raised. Rows the predicate rejects are kept in the released data with the reason attached, since they are the subject of §5 rather than attrition, and the deposit states the failure that motivated each clause.

4 Experimental Design

4.1 Research questions

Six questions organise what follows: three answered by the pre-registered family, three by evidence labelled exploratory wherever it appears. The confirmatory family is the one frozen before collection and its size stays at $m = 10$ throughout (§4.5); the exploratory questions are additions, and Table 3 says which is which and when each was formed.

RQ4 was pre-registered as an exploratory census. RQ5 and RQ6 were formed while the apparatus ran, and nothing here presents them otherwise: RQ5 after the batching rates of the collected cells were known, RQ6 after the variance decomposition of this study’s own data existed. What they rest on is not anticipation but recomputability — verbatim endpoint

responses, an arm with a stopping criterion fixed before it ran, and two calibrations a reader can recompute from the released rows.

4.2 The grid

Four models, each served by both Databricks and Amazon Bedrock under two transports: sixteen cells. The platforms are named rather than abstracted. The venue is single-blind, and the mechanisms of §5 quote the endpoints verbatim, so a reader recognises the platform from the message whatever we call it and can re-issue each probe against the platform that produced it. Every cell runs the same 45 held-out binaries eight times, so the design plans 5,760 trials while the isolated-workspace re-collection delivers 5,809 valid measurements and the original collection 5,805. The three numbers differ for one reason: a run returning an infrastructure failure is logged as a record and *re-run*, so a cell can end with more valid measurements than the 360 it planned. Every cell of both batches carries all 45 binaries; the surplus — 49 above plan in one batch and 45 in the other — is the count of runs an endpoint refused or dropped, which §5 treats as the subject rather than as attrition. One binary of one cell ends at seven valid runs of eight rather than 45 binaries of 45, for the reason given in §5.2. Turn budget, temperature and output cap are identical in every cell and recorded per row rather than assumed. The re-collection cost \$139.40 across the sixteen cells, from \$1.22 to \$21.90 per cell, and every row carries its own cost so the price of any number in this paper is auditable beside it.

Table 4: The grid. Every cell is the same corpus under a different apparatus.

models	4 (two open-weight, two proprietary)
infrastructures	2 (Databricks, Amazon Bedrock)
transports	2 (native, textual)
cells	$4 \times 2 \times 2 = 16$
binaries per cell	45, held out, frozen by hash
corpus provenance	inherited with the companion study, not drawn from a public benchmark
runs per binary	8
trials	5,760 planned, 5,809 measured
turn budget	12 exploration turns + 1 mandatory submission turn
termination	submission through the tool, or budget exhausted
temperature	0.0

4.3 Why we say *nominal* model

Throughout we write *nominal model* rather than *the same weights*. Two managed clouds advertising the same open-weights model may serve different quantisations, different runtime versions, or route to a different variant under load, and neither exposes enough to settle it — a divergence already measured directly, on 11 of 31 commercial endpoints for four Llama models (Gao et al., 2025). What we hold fixed is therefore the model *identity a user selects*, which is exactly what a published evaluation names in its table — and if two services selling one name behave differently, that difference belongs in the method section of every paper that names the model and not the service, whatever its cause.

4.4 Hypotheses, and why two of them are disjunctive

Both factors are manipulated *as deployed*: a transport is a format together with the context volume that format costs (§8), and an endpoint is a serving stack together with whatever weights it actually loads (§4.3). The hypotheses quantify over those compounds, not over idealised single-variable versions no experiment on managed clouds can isolate. We ask whether the transport moves the score at fixed infrastructure (H1), whether the infrastructure moves

it at fixed transport (H2), whether the transport’s effect is constant across models (H3), and whether the two interact (H4). An interval that excludes effects above some magnitude is not the same claim as no difference, and the second is never written where the data support only the first.

4.5 The test family, and why its size does not shrink

Ten tests were pre-registered as one family with Holm step-down, and the family size is fixed. It does not shrink when a cell turns out to be inconclusive, uninteresting or unexecutable, because removing a member after seeing the data lowers the thresholds of every survivor — a correction growing more permissive exactly as one learns which tests one would like to pass. The deliberate consequence is that contrasts nobody disputes consume correction budget, which is the price of fixing m before the data exist.

4.6 Six defaults inherited from the companion study

Six constraints are inherited from the companion study (Author(s) withheld, 2026), each traceable to a defect that produced a clean and false number: unwrapped candidate source, functions in native order, symbol tables stripped, a reasoning-only turn not recorded as a zero, a price guard that refuses *before* the provider call, and the token-limit parameter sent under the one name the endpoint accepts. None is a discovery — each is a default a competent implementer would set, which is what makes the list worth publishing — and the artifact states each with the defect it answers.

5 RQ4 (*pre-registered as exploratory*): What the Endpoint Removes Before a Score Exists

Eight of the 21 model-platform pairs we attempted never produced a score; five of the 18 on the two platforms the design measures. The third platform is Azure, probed while assembling the sample and then left out of the arms. This is the result that does not depend on a p -value — an endpoint either returned the message we quote or it did not — and it is reported at a scale a reader can size.

Three quantities are counted here, each in its own unit and with its own denominator. (A) Pairs removed before measurement: 8 of 21 attempted, spanning 11 distinct models, traceable to 7 distinct causes — six of them are the mechanisms M1–M6 below and the seventh is a reasoning-channel case the deposit carries. (B) Losses inside a cell that had already started: M7 cost 453 rows of one cell, discarded and re-collected; M8 rejected 11 trajectories across the two retained collections — four in the original collection of 5,805 valid measurements, seven in the re-collection of 5,809. (C) Distinct mechanisms documented: 8, of which 6 act before a score exists and 2 from inside a run. **Each is an existence proof with its own denominator, and no incidence rate over models or platforms is computed from them.**

The removals are not distributed evenly across the three platforms, and both denominators are therefore reported. Databricks 2 removals of 8 pairs attempted, Bedrock 3 of 10, **Azure 3 of 3**: the overall 8 of 21 becomes 5 of 18 on the two measured platforms, 27.8% against 38.1%, a difference of 10.3 points resting on three attempts against one infrastructure. Three mechanisms below were observed only there, which is why Table 5 names the platform for each. Azure stays in the denominator because dropping it would report a smaller number by discarding the attempts that failed most, and both fractions are properties of an assembled roster rather than of managed clouds (§7.5).

Table 5: The eight mechanisms, each an existence proof with its own denominator rather than a rate. **Class** is operational and says what kind of thing the removal is, which is what decides who can do anything about it: *lifecycle* is an access or retirement policy, *contract* is a constraint of the API’s own request contract, *quota* is an inconsistency between what a catalogue advertises and what the quota system holds, *adapter* is an incompatibility between the format a model emits and the validation the serving layer applies to it. **Scope** says how far the decision reaches: *vendor* holds for everyone on that platform, *region* depends on where the request originates, *tenant* depends on the account’s own history and quota, so two teams on one platform see different rosters. **Det.** is the probe that reaches the mechanism, from the matrix in the deposit: 1 an ordinary single-request capability check, 2 a deliberately constructed conversational state, 3 detectable only when the model itself emits the triggering output. Quoted cells are the endpoint’s words under the evaluated configuration, truncated to one line; M3 is a three-step control-plane observation and is set unquoted for that reason. The deposit carries every message in full with the request and the configuration that produced it.

	mechanism	platform	class	scope	det.	verbatim message
M1	protocol refusal	Databricks	contract	vendor	2	400: multi-turn tool calls unsupported
M2	deprecation	Azure	lifecycle	vendor	1	<code>ServiceModelDeprecated ... since 06/13/2026</code>
M3	absent quota	Azure	quota	tenant	1	catalogue advertises it, quota system has no row, creation refused
M4	disuse	Bedrock	lifecycle	tenant	1	<i>“marked by provider as Legacy and you have not been actively using the model”</i>
M5	jurisdiction	Bedrock	lifecycle	region	1	<i>“Access to Meta Llama models is not allowed from unsupported countries”</i>
M6	parallel calls	Azure	contract	vendor	2	400 <code>UnsupportedToolUse</code> : <i>“does not support more than one tool call”</i>
M7	tool config not expressible	Bedrock	contract	vendor	2	tool-choice admits “auto”, “any”, a named tool — not “none”
M8	endpoint rejects its own model’s output	Bedrock	adapter	vendor	3	<code>ValidationException</code> : tool name fails <code>[a-zA-Z0-9_-]+</code>

Detection level 2 means a probe exists and must be built: for M1 a history already carrying one tool call and its result; for M6 a history carrying two calls inside a single assistant message; for M7 a history carrying a tool-use block in a request that offers no tools. Each is one request once assembled. Level 3 admits no such probe: the request must carry a tool name the model itself generated.

Building the sample required probing which models each infrastructure would serve, and that probe returned six mechanisms. Two more appeared from inside the confirmatory arm, where the roster probe does not reach them: eight in total, none sought.

5.1 The eight mechanisms, with the endpoint’s own words

Every row is bounded to the configuration that produced it, listed per mechanism in the deposit: one endpoint, one API surface, one region, one set of parameters. The wording throughout is that the service rejected the request under the evaluated configuration, not that the platform does not support the feature. Where variants were tried — two identifier forms on M4, three history shapes on M6, the three admissible tool-choice values on M7 — the deposit records them, and those three rows can be stated more sharply for that reason.

Table 5 is not a taxonomy derived from documentation: every row is a message an endpoint returned to a request we made. M1–M6 were found by probing which models each infrastructure would serve; M7 and M8 appeared from inside the confirmatory arm.

The detection column is the operational finding, and it is a gradient rather than a split. Four mechanisms answer a single ordinary request. Three — M1, M6 and M7 — are invisible to any check that sends one call per turn, and become one request each only once the conversational state is assembled by hand; a harness certified healthy by an ordinary capability check can therefore still lose a cell after the money is spent. M8 admits no probe at all in this sense, because what triggers it is generated by the model during the run. The practical consequence for a pre-flight is that it must exercise realistic conversational states, and that even then one class of failure remains reachable only by collecting.

For M6 the three-request sequence below separates the levels directly, and M7 is defined by a request shape only a history containing a tool call can produce; M1’s reproducer is the agentic exchange its message names.

5.2 Commentary on M6, M7 and M8

On M6. The state it needs — a history in which the model has already emitted two calls inside one assistant message — is one no single request constructs, but it *is* constructible, and assembling it by hand turns the mechanism into a probe answerable before any collection is paid for. Its verification was three requests: one tool at turn one passes, a history with one prior tool call passes, a history with two calls in one assistant message is refused.

On M7. The protocol we pre-registered ends each trajectory with a turn that offers no tools, and under the evaluated Bedrock Converse configuration that turn *cannot be expressed*: the API requires a tool configuration whenever the history contains a tool call, and its tool-choice parameter admits “auto”, “any” and a named tool — but not “none”. We tried the three admissible values and omitting the configuration entirely; none expresses the turn.

M7 is the easiest of the three level-2 mechanisms to have found, and we did not find it that way. The constraint is in the vendor’s public API reference, which documents the tool-choice parameter as a union of exactly three members with no value for “no tools”: it was knowable before this study existed, by reading, without issuing a request. M1 and M6 have no such public statement that we could locate, and were reached by probing. The operative lesson is therefore not that M7 was undiscoverable but that *nobody checked the pre-registered protocol’s termination shape against the target API’s documented request contract* — a check that costs one page of reading and would have saved a 453-row cell.

The exposure was not where it first appeared. The broken rows carried an infrastructure-failure flag and were already excluded from every mean; what survived was structural. A trajectory reaches the final turn with a tool call in its history only if it has *not* already submitted through a tool, so the runs that completed were exactly those that had submitted early — a second selection effect, inside the axis this study exists to measure cleanly. That batch of 453 rows was discarded and the cell re-collected against a corrected client. M7 sits at the same detection level as M6 and for the same reason: an ordinary capability check reports it healthy, because it fires only in a conversational state that check does not assemble.

On M8. The open-weights model emits tool calls in its own chat format, which uses special tokens to separate reasoning channels. The serving layer does not strip that token from the tool name, so the name reaches the API as `decompile_function<|channel|>commentary`. The API then validates the name it was just handed:

```
ValidationException: Value at
'messages.10.member.content.1.member.toolUse.name'
failed to satisfy constraint: Member must satisfy
regular expression pattern: [a-zA-Z0-9_-]+
```

and rejects the entire request — not the turn, the trajectory. **That sequence is what is reported here: a name generated by the served model reaches the same service’s request validation and is rejected there.** It occurred in 11 trajectories, four in the original

collection and seven in the re-collection, all in the one cell where that model meets that endpoint under the native transport. Retrying does not reliably recover the run: on one binary the cell logged 15 attempts for 7 valid measurements, one of the eight rejected attempts carrying this exception verbatim and the other seven recording no candidate. Nothing in the released evidence identifies where in the stack the format a model emits and the validation of the service’s own API fail to meet, and it is not assigned here.

Four further observations belong to the census and not to a paper of this length, and the deposit carries each with its verbatim evidence. One carries an obligation that transfers: under a service control policy that permits invocation while denying enumeration, the roster cannot be recovered from the account and must be declared by explicit enumeration, which is what §4 does.

5.3 How far a removal moves a published comparison

That a refused model has no row to correct does not by itself say how much a reader’s conclusion changes. The census documents real cases — a Llama variant deprecated on one platform while another serves it, a second blocked by jurisdiction while its siblings run — so someone evaluating on one endpoint reads a table with rows missing and does not know which. We bound the consequence on the four models scored on both clouds, removing one at a time and recomputing the comparison the way a leaderboard reports it. The *order* survives: all pairwise orderings agree before and after every removal. The *scale* does not, and with four models its collapse is close to arithmetic — removing the lowest scorer necessarily compresses a first-to-last spread — so what these data support is the sign and the order of magnitude rather than a coefficient. The artifact carries the eight removals with their per-cloud figures.

5.4 What the census adds to what is already catalogued

The closest neighbour is a taxonomy of failure modes in multi-provider serving, classified by origin layer and detectability (Pandey and Singh, 2026). What this census adds does not depend on where any single row is filed in it, and that independence is deliberate: a reader who prefers to call M3 a configuration bug and M8 an integration bug can do so, and four things stand unchanged.

1. **Multiple distinct removal mechanisms exist and were met in one small study** — eight, none sought, each with the verbatim response of the endpoint that produced it.
2. **They sit at different points of the evaluation lifecycle.** Six act before any score exists, so the experimental unit is missing rather than mismeasured; two act inside a run that had already been paid for.
3. **Their measurement consequence is structural.** A removed pair has no row to correct, which is a selection and not a bias, and §5.3 bounds what that does to a published comparison.
4. **They require different detection.** Four answer an ordinary capability check, three require a constructed conversational state, and one is reachable only during execution — so a pre-flight that sends one request per model does not cover the class.

The question none of the existing layers asks is the one that decides whether a model appears in a benchmark at all — *will this endpoint serve this model identity, under this request contract, from this account and this region?* That is a reporting obligation before it is a taxonomic one, and it is where Table 10 puts it. There is no claim that nobody has catalogued serving failures; the artifact carries the layer-by-layer comparison so a reader can place each row wherever they judge it belongs.

Table 6: Variance decomposition. K_∞ is the number of binaries needed to resolve 3pp with unlimited runs per binary. †: on these two contrasts the components are **poorly identified at the boundary** — the estimated noise exceeds the observed variance, and the bootstrap clamps the residual to zero in 86% of replications on T7 and 69% on T8. No component is reported for them, in either direction: a clamped estimate is a boundary artefact, and printing a precise zero would report the constraint rather than the data. T1, T3 and T6 clamp in at most 1% of replications, and are the rows the decomposition is read on.

	axis	SD	% noise	K_∞
T3	transport	0.2774	0.5%	668
T6	endpoint	0.2189	9.1%	380
T4	transport	0.1409	20.6%	137
T1	transport	0.2002	46.1%	189
T2	transport	0.0916	61.0%	28
T5	endpoint	0.1292	93.1%	10
T7	endpoint	0.0117	not decomposable†	
T8	endpoint	0.0585	not decomposable†	

6 Analysis and Interpretation

The census establishes that an endpoint can remove a model before a score exists. This section asks how far the surviving numbers move, and reports beside each answer what that contrast could have separated.

All numbers here come from the isolated-workspace re-collection: an independent execution of the full $4 \times 2 \times 2$ design on workspaces isolated per cell, comprising 5,809 valid measurements. It carries the primary analysis because its apparatus removes the shared-workspace mechanism of §8 by construction, which is the reason that does not depend on either batch’s numbers. The original collection is reported beside it throughout, and the four concordance diagnostics are all met (§8). Those diagnostics corroborate rather than validate blind: their thresholds were fixed with the original collection’s outcome already known, and Appendix C states when each was written and against what.

6.1 Which lever binds is a property of the cell

The variance of a paired difference decomposes into within-binary noise, which shrinks with more runs per binary, and heterogeneity between binaries, which shrinks only with more binaries. The two are not interchangeable, and Table 6 shows the choice is not a property of the axis: for T3 and T6 more runs buy nothing, at 668 and 380 binaries against the 45 collected.

These are estimates on 45 units. Bootstrapping the binaries — the unit the contrast is paired on, 5,000 replications at a declared seed — gives T3’s noise share as 0.5% [0.0, 1.4] with K_∞ 668 [265, 1136], and T6’s as 9.1% [4.1, 21.6] with 380 [126, 744]. The interval on K_∞ is wide and its lower end is still six times the 45 binaries collected, so the conclusion that more runs buy nothing does not rest on the point estimate; neither noise-share interval reaches the 50% that would make the two components indistinguishable.

On two contrasts the components are poorly identified at the boundary, and we report none rather than a precise zero. The rows marked † have estimated noise exceeding the observed variance. That is not a finding about those cells but a variance model that does not fit them: the decomposition assumes the two conditions’ runs are independent given the binary and clamps the residual at zero when that fails. The bootstrap makes the failure countable: the residual is clamped in **86%** of replications on T7, 69% on T8 and 43% on T5. **The rule applied is that a contrast whose residual is clamped in more than half of the bootstrap replications carries no decomposition here**, and it was written

after computing the bootstrap on these eight contrasts rather than before: it separates T7 and T8 from the rest, and no conclusion of this paper rests on either. On T1, T3 and T6 clamping occurs in at most 1% of replications, which is why those three are the ones the decomposition is read on.

The two excluded contrasts are also the only two whose intervals are narrow enough for a $\pm 3\text{pp}$ boundary to fall outside them (Table 7), and that is mechanical rather than coincidental: a cell with very little observed variance yields both a narrow interval and a variance model that collapses at the boundary.

What the metric’s granularity does and does not explain

Our measurement is coarse: pass-rate over five unit tests gives a run *six* values, 53% of the 5,809 rows sit at an extreme and 33% of the 720 binary-by-cell means land exactly on 0 or 1. A floor- and ceiling-heavy measurement is on its own sufficient for a large between-binary spread, so it is a rival account of Table 6.

So it was measured on this corpus. If saturation produces the between-binary spread, removing the saturated binaries should drive the noise share up. Two exclusion rules: dropping the binaries saturated in *both* cells removes the pure artifact, while dropping those saturated in *either* cell also discards binaries one arm solved and the other did not, which *are* informative — the severe rule plays against the finding. Each share is resampled over the binaries of its own subset. On T3: 0.5% [0.0, 1.4] over all 45 binaries, 0.7% [0.0, 2.0] over the 29 surviving the first rule, 1.7% [0.0, 3.2] over the 20 surviving the second. On T6: 9.1% [4.1, 21.6], then 15.6% [10.9, 25.0], then 17.0% [11.3, 32.6]. **No upper bound reaches the half** that would make noise and heterogeneity indistinguishable, under either rule.

Excluding binaries also reduces K , so the rise could be an effect of subset size. Removing K binaries *at random* at each declared size, 20,000 times, places the observed share at the 78.7th percentile of the random distribution for T3’s first rule, the 96.0th for its second, the 96.5th for T6’s first and the 84.6th for its second. **Two of the four are individually significant at the conventional level** — the 96.0th and 96.5th percentiles, so about 4.0% and 3.5% of random subsets reach the observed share — and they are consistent with granularity contributing to the between-binary spread. The other two are not: a random subset matches or exceeds them roughly one time in five. **Conceding that contribution at its observed size changes nothing the decomposition needs:** even the largest rise, T6 from 9.1% to 17.0%, leaves the noise share far from the half at which noise and heterogeneity would be indistinguishable. The consequence for a designer is in §7.5: size the tests per task the way the tasks are sized, and report the floor and ceiling shares beside any decomposition.

6.2 RQ6 (*exploratory, arose during execution*): the calibration this design was sized on, and what it can be compared against

The pre-registration inherited a calibration of 84% run-to-run noise and $K=13$, and this design was sized on it. **Measured on this apparatus, the binding lever is the number of tasks:** 0.5% noise and a floor of 668 binaries on the contrast with the largest observed dispersion, 9.1% and 380 on the second such contrast. The inherited figure comes from a batch whose validity flag had failed, and independent reconstructions of it disagree, so what stands is the inversion measured here rather than a comparison between the two numbers.

The consequence is about where a budget goes, not how large it must be. Reaching the floor this contrast implies would take 668 binaries at eight runs on both arms — 10,688 trials, roughly \$253 at the prices its own cells paid, against the \$17 those cells cost. On contrasts with this much between-task heterogeneity, additional budget buys more from task diversity than from repeated runs of the same tasks, and the two are not interchangeable at any budget. A design sized on 84%/ $K=13$ buys the second.

The two collections agree with each other — 0.1%/705 against 0.5%/668, and 9.3%/487 against 9.1%/380 — which is **internal reproducibility**. Establishing that a variance component varies *between* apparatuses takes two apparatuses, so that question is carried to §7.5 as a proposition.

6.3 RQ1–RQ3 (*confirmatory*): a negative pre-registered result, and the resolution at which it was established

This subsection reports a *negative* pre-registered result and reports it as such: the hypotheses, the test family and the analysis were fixed before the data existed (§3), the family resolves no effect, and the outcome is stated before the exploratory findings that did produce something. What it contributes is not the absence of an effect but the *resolution* at which the absence was established, which differs by a factor of 23.7 between contrasts sharing one design.

Of the ten tests fixed in the pre-registration, *none* passes its Holm threshold at $m = 10$ (Holm, 1979). The smallest p the frozen script emits is 0.0155, against a threshold of 0.0050. Table 7 reports that frozen series and prints each test’s own threshold beside it; two sensitivity series computed after the data existed — the exact Student and a distribution-free sign-flip — give the same family outcome and are in Appendix A.

The family’s fixed composition costs something arithmetically, and the amount is computable. Two of the ten tests are not independent of the other eight, so Holm at $m = 10$ divides the smallest threshold by ten rather than by eight — 0.0050 instead of 0.0063. The smallest p in the family, 0.0155, is above both, so the outcome does not turn on that choice; on a study whose smallest p fell between them it would. $m = 10$ stands because the pre-registration fixed it and because it is the conservative direction, not as the only defensible count: correcting RQ1–RQ2 and RQ3 as separate families would also be defensible, and on these data it changes nothing, since the smallest p clears no threshold under any of those countings. The artifact carries the series.

That answers RQ1–RQ3 in the direction of “not at this resolution”, and the resolution is the reportable quantity: an evaluation that cannot separate 4pp from zero is not free to call two transports equivalent.

The pre-specified threshold and the resolvable threshold are two different quantities, and this design separates them. The pre-registration fixed $\pm 3\text{pp}$ as the band outside which an effect would falsify the null, and fixed it before any dispersion had been observed. The dispersions that arrived span a factor of 23.7 (Table 6), and the intervals of Table 7 are their direct consequence: **on six of the eight contrasts the interval is too wide for a $\pm 3\text{pp}$ boundary to fall outside it**, and on five it is wider than the 4.87pp the design was sized for. **The lesson is about sizing, not about pre-registration:** a pre-specified falsifier threshold is only usable if the planned design can resolve it, and that compatibility is checkable at design time from a variance estimate on the same apparatus (§6.2). The frozen $\pm 3\text{pp}$ boundary is accordingly read here as a pre-specified threshold and not as evidence about the contrasts whose dispersion sits above it. That sample sizes in this field are often set without such a check is not new: 103 software engineering experiments were found systematically below accepted norms (Dybå et al., 2006), a decade after the concern was raised as “missing and misunderstood” (Miller et al., 1997), and it has been documented again for language-model evaluation (Card et al., 2020).

Table 7 and Fig. 1 are the whole confirmatory result. T3 reproduces the transport difference that motivated this study at -9.83pp — different apparatus, isolated workspaces, three information leaks closed (§8) — and T6 measures 7.94pp between two deployed services serving the same nominal model at fixed transport. Neither passes its Holm threshold, and both are contrasts whose observed dispersion is among the largest in the study. All three statements hold together.

Table 7: The ten pre-registered tests, on the re-collection (isolated workspaces). **The p column is the one the frozen analysis script emits**, so this table is the pre-registered analysis and nothing else; the exact-Student and distribution-free series are sensitivity analyses and are reported in Appendix A, where the family outcome is the same under all three. δ is the paired difference per binary, over 45 binaries of 8 runs each. **Sign convention:** for transport contrasts $\delta = \text{textual} - \text{native}$, so a negative value means the textual protocol scores lower; for endpoint contrasts $\delta = \text{Bedrock} - \text{Databricks}$. The Holm column is each test’s own threshold at the pre-registered $m=10$, printed beside its p so the outcome is readable without trusting a sentence. **The interval widths carry the heterogeneity:** they differ by a factor of 23.7 across contrasts of one design, which is what makes a single pre-specified band the wrong instrument for all eight. ¶T5’s δ and interval rest on $K=45$; one of its binaries returned seven valid runs of eight on one endpoint, which the frozen script averages and the exact- p sensitivity script excludes, so that sensitivity series rests on $K=44$ for this row alone. *T6 is a contrast between two deployed services: they differ in provider-reported input tokens, in latency and in turns used (§6.7), and the source of the score difference is not identified.

	contrast	δ	95% CI	p (frozen)	Holm thr.
T6*	llama, endp.	-7.94	[-14.5, -1.4]	0.0155	0.0050
T3	haiku, transp.	-9.83	[-18.2, -1.5]	0.0181	0.0056
T1	gpt-oss, transp.	-6.56	[-12.6, -0.5]	0.0290	0.0063
T5¶	gpt-oss, endp.	+3.26	[-0.6, +7.1]	0.0893	0.0071
T9	heterogeneity	—	—	0.1011	0.0083
T2	llama, transp.	+2.22	[-0.5, +5.0]	0.1055	0.0100
T7	haiku, endp.	+0.22	[-0.1, +0.6]	0.2054	0.0125
T8	sonnet, endp.	-0.94	[-2.7, +0.8]	0.2812	0.0167
T10	interaction	-4.16	—	0.3448	0.0250
T4	sonnet, transp.	-1.00	[-5.2, +3.2]	0.6358	0.0500

6.4 RQ1–RQ3 (*confirmatory*): what a contrast could separate is a property of the cell

The pre-registration sized the design with a pooled standard deviation of 0.1062 and a conservative 0.1167, both calibrated on the companion study, giving one minimum detectable effect of 4.87pp for every cell. The observed standard deviations of the paired differences span 0.0117 to 0.2774, **a ratio of 23.7**, and the interval widths of Table 7 are the same fact in the units of the effect: they are the standard deviations rescaled by a constant, so the heterogeneity is readable off either column and needs no third. What it means is that a design sized on a single band quotes one number for eight contrasts that differ by more than an order of magnitude in what they can separate — which is a property of the cells, not of the study, and is the reason §6.2 treats the sizing axis as an empirical question.

A null result here therefore reads: no contrast places its effect outside the band its own dispersion permits. It does *not* read as equivalence — $K = 45$ does not support an equivalence test at 3pp, which would need 99–119 binaries, and this was declared before the data existed.

6.5 RQ5 (*exploratory, arose during execution*): how much of the transport effect call batching could account for

The pre-registered transport contrast moves two things at once on the models that batch: the format of the call and the number a turn admits — haiku issues 1.438 calls per turn against sonnet’s 1.333. A declared confound is not a bounded one, so the arm that separates them was collected: native transport forced to *one call per turn*, all else held fixed. It is exploratory, does not enter the family of ten, and its stopping criterion was fixed before it ran.

Removing batching does not lower the score: both point estimates are *positive* and both intervals contain zero (Table 8), against a transport effect of -9.83pp on the same model and

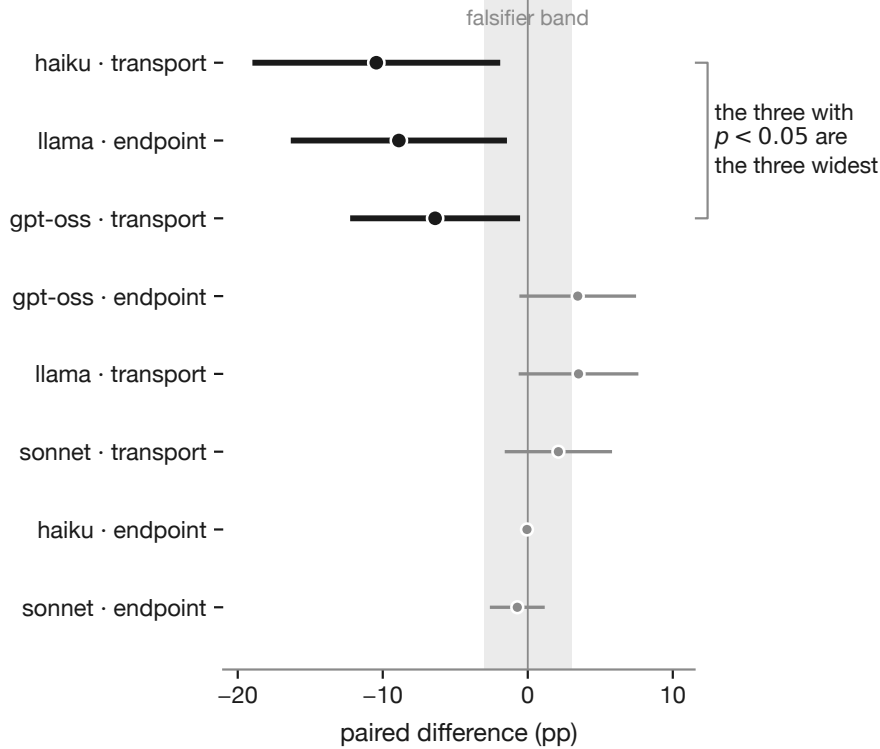


Figure 1: The ten contrasts, ordered by p . Filled and heavier lines are the three below 0.05 uncorrected; none passes its Holm threshold. Those same three carry the widest intervals in the study, so the ordering by p and the ordering by precision run opposite to one another. Dotted lines mark the pre-registered $\pm 3\text{pp}$ falsifier band, which six of the eight contrasts could not have resolved (§6).

Table 8: Constraining the native transport to one call per turn. Paired on binaries, $K = 45$, eight runs per binary. The arm is exploratory and does not enter the corrected family.

model	full	constr.	diff.	95% CI
haiku-4-5	0.747	0.754	+0.8pp	$[-2.6, +4.2]$
sonnet-4-5	0.793	0.809	+1.6pp	$[-1.3, +4.4]$

endpoint. **The arm bounds rather than eliminates, and the bound is arithmetic.** Read at the most unfavourable end of each interval — -2.6pp on haiku, -1.3pp on sonnet — batching would account for at most **26%** and **13%** of the -9.83pp . That is what the arm supports: not that batching contributes nothing, but that it is not the principal account, with a ceiling attached.

6.6 RQ5, second finding (*exploratory*): one refusal reconfigures the model for the rest of the run

The arm produced a second result. Across 1,084 trajectories *every* first turn issues more than one call and is constrained; after the first refusal the rate collapses to 0.2%, eight turns out of 4,457. The model does not learn the constraint from the tool schema, which does not encode it — it learns it from one rejection, and the lesson holds for the rest of the trajectory.

The arm falls outside its pre-declared validity band, and that governs what it licenses. The band required 33–38% of turns to carry a constrained call, estimated from

Table 9: T6’s two cells, native transport, 360 runs each. Median with first–third quartile; the ratio is Bedrock over Databricks. Corpus, turn budget, temperature and output cap are identical by construction. *Turns used* is counted from the per-turn traces, one record per turn, and is not the configured budget: the budget is 12 exploration turns plus a mandatory submission turn in both cells. Input and output token counts are the ones *the provider reports*; see the text for what that does and does not license.

	Databricks	Bedrock	ratio
provider-reported input tokens per run	20,562	8,853	0.43
latency per run (s)	13.4	8.6	0.64
turns used (median)	6.0	4.0	0.67
provider-reported output tokens per run	448	303	0.68
candidate characters	598	544	0.91
output tokens per second	33.2	35.1	1.06
runs reaching the turn budget	0.28%	0.00%	—

the unconstrained batching rate; the observed figure is 19.7%. Under the rule fixed in §4 the arm therefore licenses no statement about the *size* of the batching effect, and none is made. The reason the band was missed is itself the measurement above: it assumed a batching rate constant along the trajectory, whereas the collapse fixes the achievable share at essentially one constrained turn per trajectory. Of the 5,541 turns carrying a call, 1,084 are first turns and all are constrained, and eight more are constrained later — 1,092 in total, **19.7%**, which is the structural ceiling of 19.6% plus those eight. The band’s 33–38% was unreachable under the constraint, so its premise was wrong rather than the data.

The arm therefore licenses the ceiling above and no point estimate of the batching effect, and the ceiling needs only the two intervals rather than the band that was missed. It also explains an asymmetry in the main comparison — under the prompted-text protocol the option to batch never exists, while under constrained native it exists for one turn and the model abandons it after one refusal.

6.7 What differs between two deployed services serving one model identity

T6 measures 7.94pp between two services serving the same model identifier, which §4.3 defines as a nominal identity rather than a claim about weights. What follows identifies no component as the source of the score difference. It reports that the two services differ measurably in several quantities a serving stack determines, so the contrast is between deployed services rather than between one model under two labels.

The input-token divergence is a provider-reported accounting difference, and it decomposes. Per run, the two services report a ratio of 0.43. Two components of that are visible in the released traces. The first is trajectory length: the median run takes six turns on one service and four on the other, so more history accumulates before the run ends. The second is what the counters report at the *first* turn, where no history exists yet and the client sends the same system prompt, the same four tool schemas and the same initial message: there the reported counts are 1,263 against 678, a ratio of 0.54. **Two readings of that ratio are not separated by the released evidence:** the two stacks may serialise the same content differently — the tool schemas travel in each provider’s own format — or they may count the same payload differently. Both are figures the providers report, and the released traces record exactly that: per-turn provider usage. **The quantity is therefore a provider-reported input-token accounting difference**, which is what the two services disagree about, and the paper uses it no further.

The source of T6’s score difference is not identified here. Output throughput agrees to within 6% while several other quantities do not, and that pattern is consistent with

more than one account; naming one would be an inference the design does not license. What stands is narrower and sufficient for the paper’s argument: the two deployed services differ in trajectory length, in latency and in reported context volume, at fixed nominal model identity, so **T6 is a difference between deployed services** and no design that names only the model can attribute it further. That is the same form the transport axis takes in §8. The companion study identified context volume as a free parameter in its own right (Author(s) withheld, 2026); here it is one of several quantities that move together with the factor being manipulated, and Table 7 should be read with that attached to T6.

The turn budget, measured rather than assumed

Because Table 9 raises it, the budget is reported across the whole design rather than for one cell. The re-collection produced 5,880 trajectories, of which 5,809 are valid measurements by the predicate of §3. Counting trajectories, **68 reach the turn budget — 1.16% —** and all 68 are in the native arm; no textual-transport trajectory reaches it. The saturation is localised rather than widespread: it is concentrated in the two `gpt-oss-120b` native cells, at 10.0% on Databricks and 6.3% on Bedrock, and is at or below 0.3% in the other fourteen. Median turns used per cell range from 3 to 6 against a budget of 12. Two consequences follow and neither is speculative. Every effect reported here is conditional on this fixed budget, and the budget binds asymmetrically: it binds only where a model can batch calls, which is the arm this study manipulates. The turn budget and the termination condition are therefore fields in Table 10, not footnotes.

6.8 Temperature zero is not determinism, and one demonstrated case where it differs by service

Two properties of the apparatus are usually assumed and turn out to be measurable. *First, temperature zero is not determinism.* All cells ran at temperature 0.0, and across the eight runs of the same binary in the same cell haiku returns an identical pass-rate on 91–100% of binaries against gpt-oss’s 20–24%, with 56% of binaries over the study yielding eight identical runs (Fig. 2).

Second, **one comparison of eight demonstrates that this stability can differ between two deployed services at fixed nominal model identity.** On 45 binaries the sampling separates in exactly one case: llama on the native transport returns identical scores on 24.4% of binaries at one endpoint [14.2, 38.7] and 66.7% at the other [52.1, 78.6], Wilson bands not touching — and they remain disjoint at the Bonferroni-corrected level for eight comparisons, [11.5, 44.7] against [46.3, 82.2]. The other seven overlap and are not used as evidence. **This is an existence demonstration and it is not read here as a population claim:** it establishes that score stability need not be a property of the model identity alone, not that stability generally differs between clouds, and one case out of eight cannot support the second. The comparison falls on the same model and axis where T6 shows the largest endpoint effect in the study, and the original collection isolates the same comparison.

6.9 What the metric buys, measured

Pass-rate has a floor reachable without reconstructing anything, and it was measured rather than assumed: the 53 runs that called *no* tool average 0.1132 against 0.6378 for the 5,756 that called at least one. The floor exists and does not dominate — the metric is a lower bound with a measured floor, and the artifact carries the distribution.

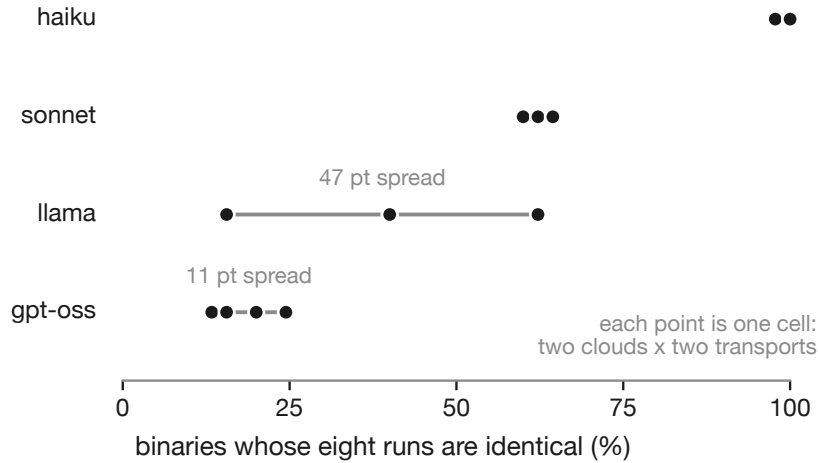


Figure 2: Determinism at temperature 0.0: the share of binaries whose eight runs return an identical pass-rate, per model, with one point per cell. The spread within a model is the distance between its clouds and transports — the same nominal model is not equally stable on different infrastructures.

6.10 Declared covariates

Two confounds were measured before the confirmatory numbers were read, and neither is corrected for: doing so after the fact is the freedom pre-registration exists to remove. They are calls per turn, which the ablation arm of §6.5 isolates and which binds on two models of four, and material acquired, where the native arm asks for more functions per binary in all eight cells. The deposit carries both with their per-cell figures.

6.11 Scope

The effect sizes are circumscribed twice over and both bounds are part of the result. They come from *one* family of tasks, decompiled C binaries scored by unit tests, on four models and two clouds, so T3’s -9.83pp is a measurement on this corpus rather than a quantity transferable elsewhere: a reader who needs a number for their own setting should take the method and the variance decomposition, not the point estimate. **And every empirical statement above concerns *our prompted-text protocol*** — one call per turn, on the last line of the response — **rather than textual tool calling in general**, about which a leaderboard reporting both modes finds the prompted variant *ahead* by 6pp on one model, against the direction that seeded this work. Nothing here is evidence about prompted-text protocols that batch calls, that use a different delimiter, or that return results in a different role.

7 Discussion

7.1 Why the census carries the weight

The transport moves a number, which is a bias. The endpoint deletes a row, which is a selection. The two also differ in what it takes to check them: the census is checked by re-issuing a probe, the contrasts by re-running 5,809 trials. **Nothing in §5 is an estimate** — an endpoint returned that message or it did not, and the artifact carries the request and the configuration that produced it — while every number in §6 is a point estimate with an interval a larger study could move.

The distinction that matters for reading §6 is between a *null* result and an *uninformative* one, and what separates them is stated per contrast rather than for the design: the effect each contrast could have resolved given its own dispersion ranges over a factor of 23.7. A design sized on a single band would have quoted one number for eight contrasts that differ by more than an order of magnitude in what they could separate.

7.2 What the null confirmatory family could resolve

Ten pre-registered tests, none passing its threshold, and the sizing that produced them is locatable. §6.2 measures the lever that binds on this apparatus at the number of tasks — 0.5% run-to-run noise and a floor of 668 binaries on the contrast with the largest observed dispersion — against the 84% and thirteen the design was sized on.

The consequence for a designer is that the sizing axis is itself an empirical question, answerable only on the apparatus that will be used. A calibration inherited across apparatuses can send a budget to repeated runs when what binds is task diversity, and no amount of care at design time substitutes for measuring it. The rival account for the decomposition — that the granularity of the metric produces the spread on its own — is measured rather than argued: §6.1 recomputes the decomposition on the subsets where the metric is at neither floor nor ceiling.

7.3 When to discard a batch, and when not to

Two batches were invalidated and re-collected rather than patched, and the rule separating them from the one kept is reusable. **Discard when the contaminating channel is systematic and asymmetric between the compared arms; report and bound it otherwise.**

The larger invalidation — 2,390 measurements — meets that test: the algorithm’s name reached the model through three separate channels and the two arms encountered one at different rates, so left in place it would have turned part of the transport effect into a measure of how often each transport stumbled onto the answer. No downstream analysis separates a contaminant that correlates with the manipulated variable.

The shared working directory (§8) fails the same test in the other direction: non-directional, thirty times in the original batch’s 5,805 measurements, with no asymmetry between arms. So it was bounded, probed, and re-collected on isolated workspaces rather than discarded. Both batches remain in the artifact with their recomputation.

7.4 What an evaluation should report

What follows amends an existing checklist rather than proposing a new one: the community guidelines already require the harness to be described once it goes beyond bare API calls (Baltes et al., 2026), and each field below either sharpens that requirement or makes explicit something those guidelines leave implicit. Each costs a line in a method section, and Table 10 states them as fields because a recommendation phrased as prose is one a reader agrees with and does not apply. The first two carry the argument of §6.7: a model name does not identify a measurable system, two deployed services under one model identity differing in trajectory length, latency and reported context volume.

The study’s own values are in §4 and §3. The tool row admits two ways of being met, and this deposit takes the stronger one: rather than naming a decompiler version a reader would then have to obtain, it ships the decompilation outputs themselves, frozen and hashed — which is what the harness consumes, so every trial reproduces without resolving a version at all.

Table 10: **What to declare in the method section of an agentic evaluation.** Eight fields, each one line, enough to reconstruct the interaction apparatus. The first six sharpen or make explicit what the guidelines we extend already require of a harness description (Baltes et al., 2026); the last two are not explicit in that checklist. The comparison is against that source alone. This table is meant to be copied into a method section as it stands, and it is the transferable output of this study together with the variance-decomposition method of §6.1.

field	what it means
transport	how a tool call is encoded — native function calling, a prompted-text protocol, typed code stubs — <i>and</i> whether more than one call per turn is admitted, since that is a second manipulation travelling inside the first
endpoint and API surface	which service served the weights, under which API and version, from which region; not only the model name
model identifier	the string the endpoint accepts, including any regional or inference-profile prefix, which can change the response to an otherwise identical request
decoding configuration	temperature, output cap, context limit, and seed where the endpoint honours one
interaction budget	the maximum number of turns <i>and</i> the condition that ends a trajectory, since effects are conditional on both — validated against the target API’s documented request contract (tool-choice values, multiple calls per turn) <i>before</i> the protocol is frozen, since a protocol that the endpoint cannot express costs a whole cell
tools and their provenance	the tools offered, their schema, and the version of any external analysis tool they wrap — or, where that version is not recorded, the frozen tool outputs released in its place
models excluded reason for each	which models were considered and not measured the endpoint’s own message, verbatim, together with the configuration under which it was returned

7.5 Seven propositions that should outlive this task

Within the scope §8 bounds, seven propositions about how experiments of this kind behave, each instantiated by this study rather than proved in general.

1. The transport belongs beside temperature in a method section.
2. An endpoint can produce missing experimental units, whose cost §5.3 bounds.
3. Variance structure differed substantially across the contrasts of this study, so a single inherited calibration is unreliable across the cells of one design. What follows for a designer is an independent pilot calibration, or an internal re-estimation whose rule is specified in advance and read blind to the outcome — not a recalibration decided once the first outcome-bearing cell is visible, which would reintroduce the design freedom the pre-registration removes.
4. A pre-flight must exercise realistic conversational states: three of our eight mechanisms are invisible to a probe that sends one call per turn, and a fourth is reachable only during execution.
5. Metric granularity limits what a variance decomposition can say. We report the floor and ceiling shares beside ours, recompute the decomposition on the subset where the metric is at neither (§6.1), and did not size the tests per task.
6. An incidence rate for pre-score removal requires a *sampled* roster, not an assembled one. Ours is assembled — the models two clouds happened to serve, plus those probed while

building the sample — so it yields the fraction inside itself, 8 of 21 pairs, and no rate beyond. The measure this study does not supply is a probe run against a roster drawn by a stated sampling rule, which would turn eight existence proofs into an estimate with an interval.

7. Whether a variance component is specific to an apparatus is not decidable from one apparatus. Two collections of ours agree with each other, which is reproducibility and not transfer; the question needs two apparatuses whose calibrations are both verifiable, and the one we inherited was not.

The artifact carries each with its argument.

8 Threats to Validity

Each threat is reported in the same four parts: what it is, how large it is, the diagnostic or mitigation that answers it, and the residual.

8.1 Construct validity: what pass-rate measures

Threat. Pass-rate is a lower bound on semantic equivalence, and it has a floor: a candidate written without consulting the decompiled program can still pass some tests, so part of any score may be pattern-matching rather than reconstruction.

Quantification. The floor is measurable inside the study rather than assumed, and §6 reports it: the runs that called no tool at all average 0.113 against 0.638 for those that called at least one.

Residual. The floor exists and does not dominate, and both halves carry a number. Pass-rate is reported as a lower bound throughout, and no claim in this paper requires it to be more.

8.2 Construct validity: what a transport carries

Threat. The transport contrast is a change of *format* and also a change in the volume of context the request carries. §6.7 reports the same quantity between two services; it also differs between the two transports, on the axis this study exists to isolate.

Quantification. Median provider-reported input tokens per run, native over textual, across the eight cells where both arms exist: 2.89 and 2.47 on gpt-oss (Databricks, Bedrock), 2.34 and 1.02 on llama, 1.38 and 1.37 on haiku, and 0.75 and 0.73 on sonnet. The native arm is reported higher in six cells of eight and lower in two, so the divergence has no fixed direction. T1 is the cell where this matters most: gpt-oss issues exactly one call per turn, so it is the contrast with no batching confound to absorb the difference.

Diagnostic. This is a component of the transport as evaluated, not a leak into it. Under native calling the tool schemas travel in a `tools` field and results return in the provider’s tool role; under our prompted-text protocol both travel as ordinary messages. **The two implementations we evaluated do not isolate serialisation format from context volume**, and the ablation arm of §6.5 separates batching from format but not format from volume. Whether some other pairing of implementations could hold context volume fixed while changing the encoding is a question this design does not answer and does not foreclose.

Residual. The transport effects of §6 are therefore effects of a transport as deployed rather than of a serialisation syntax holding all else equal, and a reader sizing their own study should read them that way.

Table 11: The same eight contrasts measured twice: the original collection, and the isolated-workspace re-collection that carries the primary analysis. Differences in percentage points; the last column states whether the re-collected effect falls inside the original 95% interval. The four thresholds are **pre-specified concordance diagnostics**, not a pre-registered rule: §8 states when each was written and against what was already known. What licenses the re-collection as the primary basis is not these diagnostics but the apparatus: the shared-workspace mechanism is removed by construction.

	model	orig.	re-coll.	in CI
T1	gpt-oss-120b	−6.4	−6.6	yes
T2	llama-3.3-70b	+3.5	+2.2	yes
T3	claude-haiku-4-5	−10.4	−9.8	yes
T4	claude-sonnet-4-5	+2.1	−1.0	yes
T5	gpt-oss-120b	+3.4	+3.3	yes
T6	llama-3.3-70b	−8.9	−7.9	yes
T7	claude-haiku-4-5	−0.1	+0.2	no
T8	claude-sonnet-4-5	−0.7	−0.9	yes

8.3 Construct validity: what the ablation arm separates

Threat. The constrained-native arm of §6.5 separates call batching from call format only up to the two effects below.

Quantification. The constraint does not remove batching alone: forcing one call per turn also changes the shape of the trajectory the model plans, so what the arm bounds is an upper limit on the batching contribution rather than an estimate of it. And it covers one infrastructure and the two models that batch at all — on the other two the rate is exactly 1.000 calls per turn, so there is nothing to remove and no contrast to run.

Residual. The arm therefore supports a ceiling rather than a point estimate — at most 26% and 13% of the transport effect on the two models that batch (§6.5) — and the artifact carries both bounds with the per-model rates.

8.4 Internal validity: a working directory shared across cells

Threat. Candidates were compiled in a directory keyed on the program and the run index and nothing else, so under concurrent collection two cells could share a compilation path and a run compiling another cell’s candidate would be scored normally with no flag raised. The key omits the cell, so no contrast is structurally exempt.

Quantification. The precondition — two cells sharing a compilation path at the same moment — arose **thirty** times in the original batch of 5,805 measurements. The deposit carries the mechanism, the write-to-read window and the probe these data admit.

Mitigation. The working directory now carries the cell, and the whole design was collected again on workspaces isolated per cell. **That is what makes the re-collection the primary basis: the mechanism is absent there by construction, not observed absent** — thirty occurrences originally, zero under 451 concurrent measurements that would have exposed it. This is a property of the code rather than of either batch’s numbers.

Concordance with the original collection. Four diagnostics were specified for the comparison, and all four are met: six of eight contrasts keep their sign against a threshold of six; no test passes its Holm threshold in either batch; seven of eight re-collected effects fall inside the original 95% interval, the exception being T7, whose original interval spans 0.3pp because that cell is almost noiseless; and the noise share that inverted the sample-size calibration reproduces within an order of magnitude, $0.1\%/K_\infty=705$ against $0.5\%/668$. Appendix C states when each threshold was written and against what was known.

Residual. The original batch is reported in full beside the primary one rather than invalidated: the criterion this programme applies to invalidation is a systematic, arm-asymmetric channel (§7), and the shared workspace is neither.

8.5 Conclusion validity: the guards, and their perimeter

Threat and residual. The analysis is protected by a content-hash chain over six named files — the pre-registration, the frozen analysis and power scripts, the held-out corpus, the collector, and the comparison script — and by an append-only convention on the result directory whose guarantee is the committed history. Both perimeters are stated so that a claim of precedence can be checked against the right one, and the artifact enumerates them.

8.6 Conclusion validity: the estimator

Threat. The pre-registered test is a paired t over per-binary means, which discards the within-binary structure a run-level model would use.

Quantification. What we can measure is where the variance sits, and Table 6 measures it: on T3 and T6, the two contrasts on which the variance decomposition is most informative, run-to-run noise is 0.5% and 9.1% of the paired variance, so almost all of it is between binaries rather than within them. On T5 it is 93%, and on T7 and T8 the estimator does not fit at all.

Diagnostic. The run-level counterfactual is computed in Appendix B and agrees: a marginal model with cluster-robust errors reproduces the frozen test’s reading on all three contrasts it covers, and none passes Holm.

Residual. That model is reported as a sensitivity analysis rather than substituted for the frozen estimator: choosing an estimator after seeing which contrasts it favours is the freedom the freeze exists to remove.

8.7 External validity: what the design does not settle

Threat. The design manipulates transport and infrastructure at a fixed *nominal model identity*, and the weaker phrase is deliberate: the design does not verify that a nominally identical model is byte-identical across two providers, and neither provider exposes enough to settle it.

Quantification. The divergence matters rather than being hypothetical: on the one axis where the Wilson bands separate, the same nominal model returns identical scores on 24.4% of binaries at one cloud against 66.7% at the other (§6).

Residual. The rates are reported at the level the design identifies — the deployed service — since attributing them further would require knowing what each provider serves. The scope of every effect size reported here is one task family, four models and two clouds, as the pre-registration bounds it.

9 Conclusion

A serving endpoint decides which models an evaluation can measure, and it decides before any score exists. Six mechanisms remove a model-platform pair outright and two destroy measurements from inside a run that has already been paid for, each documented with the endpoint’s own words and the configuration that produced it. Eight of the 21 pairs attempted here never produced a score.

They are not equally easy to find, and that gradient is what a practitioner needs. Four answer an ordinary single-request capability check. Three are invisible to any check that sends one call per turn and require a conversational state assembled by hand. One appears only while the model is generating. A pre-flight built from single requests therefore does not cover the class, and a harness certified healthy can still lose a cell after the money is

spent — which is why the last two fields of Table 10 ask for the models an endpoint refused and the message it refused them with. Where each mechanism is filed in an existing taxonomy changes none of this: what the census establishes is that several such mechanisms exist, that they act at different points of the evaluation lifecycle, that a removed pair leaves no row to correct, and that they demand different detection.

A property of the apparatus usually assumed turned out to be measurable and to vary: temperature zero is not determinism, with one model returning identical runs on 91–100% of binaries and another on 20–24%. That output distributions can differ between services at one model identity is already established (Gao et al., 2025); in one comparison of eight the difference here reaches the stability of the score itself, 24.4% of tasks against 66.7% — an existence result, not a claim about clouds in general.

The pre-registered family answers the magnitude question at the resolution the design reached. Four models, two managed clouds and two tool transports were run against hypotheses and an analysis script frozen together before any data existed, with two of the ten tests implemented under a timestamped amendment (§3). No test passes its Holm threshold, and two sensitivity series computed afterwards agree. What the family adds is the resolution itself: across eight contrasts of one design, the effect each could separate spans a factor of 23.7, so what an experiment of this kind can resolve is a property of the cell and not of the study. On the two contrasts with the largest between-task dispersion the lever that binds is the number of tasks rather than the number of runs — 0.5% run-to-run noise and a floor of 668 binaries — so a budget sized on a borrowed calibration goes to the axis that does not bind. That is a statement about what a design can resolve, and it is separate from the family’s outcome on effect sizes.

The transport moves a number, which is a bias. The endpoint deletes a row, which is a selection. The first is bounded, correctable, and already visible in a public leaderboard; the second is not, because the model it refused has no line to correct. Existing reporting guidance does not make all of the apparatus fields examined here equally explicit. Declaring the transport, the endpoint, and the models each endpoint refused costs three lines in a method section, and without them a published agentic number does not identify the system it describes.

Declarations

Funding

This work received no specific grant from any funding agency. Inference costs were borne by the author’s institution and are itemised per arm in the replication package, so the cost of each number is auditable beside the number.

Ethical approval

Not applicable. The study involves no human participants and no personal data. It measures commercial inference endpoints under their ordinary terms of use, at a request rate below the published limits of both platforms, and reports the responses those endpoints returned.

Informed consent

Not applicable. The study has no human participants. The subjects of measurement are commercial inference endpoints, and what was collected from them is the text they returned to documented requests.

Author contributions

Lorenzo De Tomasi is the sole author and carried out the whole of the work: conceptualisation, the pre-registration and its frozen analysis plan, the harness for both transports, data collection across both clouds, the analysis scripts, the interpretation, and the writing of the manuscript. There are no other authors.

Data availability statement

The replication package contains the 45 stripped binaries and their test suites, the decompilations, the harness with both transports, every raw result row including the two invalidated batches, the per-turn session traces, the frozen pre-registration and analysis script with their content hashes, and the dated succession documents accounting for every change to a frozen file. Each number in this paper is produced by a script released with it, re-running from the released rows; no figure is drawn by hand.

The two numbers this paper takes from the companion study are re-runnable here: the deposit carries a snapshot of exactly the cells they need with their per-turn logs, so both the calibration inversion and the tool-exposure asymmetry recompute without leaving the artifact. The rest of that study is not part of this deposit.

Three commands regenerate everything a reader might want to check, and they are the ones we ran:

```
python3 analysis/numeri_paper.py — every number in prose
python3 analysis/tabella_principale.py — Table 7
python3 analysis/tabella_principale.py --varianza — Table 6
python3 analysis/conteggio_m8.py — the trajectories M8 rejected, per batch
```

The first carries a known-answer check: pointed at the original collection it must reproduce the numbers this paper reported before the re-collection, and exits non-zero otherwise. The deposit’s README states how each collection is selected and what a partial selection returns.

The package is openly available at <https://github.com/lodetomasi/transport-endpoint-agent-ic-eval-artifact> and will be deposited on a permanent archive with a DOI on acceptance. Cloud workspace and profile names, absolute paths and one internal account identifier are replaced by placeholders, and the deposit carries a substitution table stating every one by its reason; nothing else is redacted, the venue being single-blind. Because redaction changes the content hash of a frozen file and this paper quotes those hashes, the deposit carries its own recomputed table with the integrity guard run and passing inside it. Two properties are checkable rather than asserted: the result directory is append-only, with invalidated batches moved and annotated with their recomputation rather than deleted; and every change to a frozen file carries a document written before the change, declaring the previous and the new hash, with a guard rejecting any divergence no document accounts for.

Conflict of interest

The author declares no competing interests. The two serving platforms measured in this study were selected because the institution already held accounts on both; neither vendor was contacted, informed, or involved at any stage, and neither reviewed this manuscript.

Reporting checklist

The study follows the community guidelines for empirical studies involving language models (Baltes et al., 2026): model identifiers, versions and configuration are reported per cell (§3); the system and harness design is described and released; per-turn session traces are published

for every run rather than summarised; and two of the four models under evaluation are open-weight, so that their scores can be independently re-run on infrastructure the reader controls. The guidelines’ requirement that we sharpen — reporting the transport — is the subject of the paper rather than only its practice.

Use of language models in this work

Language models are the object of study here, and they were also used as a tool while conducting it — to draft and review code and prose — which the community guidelines ask to be declared separately from their role as subject (Baltes et al., 2026). No measurement, no number and no decision reported in this paper was produced by that use: every quantity comes from a committed script run against committed rows.

A Distribution-free sensitivity for the pre-registered family

The pre-registered test is a paired t over 45 binaries, which assumes approximately normal paired differences. On this metric they are not: pass-rate over five unit tests takes six values, 53% of rows sit at an extreme, and the paired difference of two such means is discrete with short tails. If a test assuming no distribution reaches the same conclusion, the assumption was not carrying the result. For paired differences the exact permutation test is the sign flip — under the null the sign of each binary’s difference is exchangeable — and with 2^{45} assignments we sample 50,000 at a declared seed and report the Monte Carlo error of the p itself, a sampled p being an estimate.

Table 12: **Two sensitivity series beside the pre-registered one.** The *frozen* column is the p the pre-registered script emits and is the series Table 7 reports; the other two were computed after the data existed and are sensitivity analyses, not confirmatory results. *Exact* is the exact Student series on the same paired quantities; *sign-flip* is the distribution-free permutation on 50,000 sampled sign assignments at a declared seed. Holm applies the same fixed $m=10$ throughout. The Monte Carlo column is the standard error of the sampled p itself: a p obtained by sampling is an estimate, and reporting it without its uncertainty would be the error this paper objects to elsewhere. No test passes its Holm threshold under any of the three series, so the family outcome does not depend on which one is read.

	p frozen	p exact	p sign-flip	MC error	exact $\rightarrow$ flip
T6	0.0155	0.0191	0.0137	0.0005	−0.0053
T3	0.0181	0.0218	0.0223	0.0007	+0.0005
T1	0.0290	0.0334	0.0332	0.0008	−0.0002
T5	0.0893	0.1159	0.1206	0.0015	+0.0047
T2	0.1055	0.1107	0.1214	0.0015	+0.0107
T7	0.2054	0.2095	0.4999	0.0022	+0.2904
T8	0.2812	0.2844	0.3232	0.0021	+0.0388
T4	0.6358	0.6363	0.7751	0.0019	+0.1388

No test passes Holm under any of the three series, so the family outcome does not depend on the distributional assumption, and nothing pre-registered changes.

Two contrasts disagree by more than the Monte Carlo error, informatively: T7 moves from 0.21 to 0.50 and T4 from 0.64 to 0.78, and both are the contrasts with the smallest observed variance, where the t divides by a standard error the permutation test never uses. That the t reports smaller p exactly where variance is smallest is the mechanism rather than a coincidence, and it supports the reading of §6: T7’s narrow interval reflects how little that cell varies, not how much evidence it carries. This analysis is a sensitivity result and does not replace the

frozen one: had the series diverged on the family outcome, the reportable conclusion would be the pre-registered one with the divergence stated beside it.

B A run-level model, and what it changes

The pre-registered test operates on per-binary means. A reader may ask what a model at run level would have found, and the released rows allow the answer to be computed rather than argued. A marginal binomial model with cluster-robust errors — GEE with exchangeable working correlation, clustered on the binary, over each contrast’s individual test outcomes — gives $z = -2.43$ ($p=0.0152$) on T3, $z = -2.45$ ($p=0.0142$) on T6, and $z=1.63$ ($p=0.1039$) on T5. That is the frozen test’s reading: the same two nominal significances, the same non-significance, and nothing surviving Holm at $m=10$. The run-level structure changes no conclusion.

The same script also fits a variational binomial mixed model, and we report it as a caution rather than a result: it returns $|z|$ between 5 and 17 on the same contrasts where the paired t and the GEE both give $|z|$ between 1.6 and 2.5, which is the signature of a variational approximation understating posterior uncertainty, not of an effect the design missed. Both fits are exploratory, enter no family, and their reading criterion was written into the script before the numbers existed.

C Provenance of the re-collection’s concordance diagnostics

§8 grounds the re-collection’s role in the apparatus: the shared-workspace mechanism is absent there by construction. The four concordance diagnostics are a separate instrument, and this appendix records their provenance so that a reader auditing the artifact finds the timings already stated rather than having to reconstruct them. Every figure recomputes from the released rows.

Table 13: What existed when each artefact carrying the four thresholds was written. Shares are of the 5,880 rows the re-collection eventually produced, which yield the 5,809 valid measurements reported elsewhere. “Contrasts computable” counts how many of the eight paired contrasts had both their cells complete at that moment. All three counts are reconstructed from the timestamps carried by the released rows, at the exact second of each event; the git history is what dates the events, not what supplies the counts.

artefact and timestamp	rows	share	contrasts computable
analysis script, file timestamp	112	1.9%	0 of 8
amendment prose, file timestamp	1,611	27.4%	2 of 8
both, entering version control	4,318	73.4%	5 of 8

Two readings follow. Relative to the re-collection, the thresholds were fixed while the batch had produced one cell of sixteen from a single model, and six of the eight paired contrasts remained uncomputable even at the latest of the three timestamps. Relative to the original collection, which had completed, they were fixed with its outcome known. The four are accordingly used as **pre-specified concordance diagnostics on a comparison**, and the choice of primary basis rests on the apparatus (§8), where the contaminating mechanism is absent by construction.

References

Why does function calling lower model accuracy on Berkeley Leaderboard? discussion #606, gorilla repository. <https://github.com/ShishirPatil/gorilla/discussions/606>.

- Berkeley function-calling leaderboard, version v4. <https://gorilla.cs.berkeley.edu/leaderboard.html>, 2026. Modes defined verbatim on the leaderboard: “FC = native support for function/tool calling. Prompt = walk-around for function calling.” Accessed 15 August 2026.
- Casper Albers and Daniël Lakens. When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of Experimental Social Psychology*, 74:187–195, 2018.
- Florian Angermeir, Maximilian Amougou, Mark Kreitz, Andreas Bauer, Matthias Linhuber, Davide Fucci, Fabiola Moyón C., Daniel Mendez, and Tony Gorschek. Reflections on the reproducibility of commercial LLM performance in empirical software engineering studies. In *Proc. 48th Int. Conf. on Software Engineering (ICSE)*, 2026.
- Berk Atil, Sarp Aykent, Alexa Chittams, Lisheng Fu, Rebecca J. Passonneau, Evan Radcliffe, Guru Rajan Rajagopal, Adam Sloan, Tomasz Tudrej, Ferhan Ture, Zhe Wu, Lixinyu Xu, and Breck Baldwin. Non-determinism of “deterministic” LLM settings, 2024.
- Author(s) withheld. How much of an agent’s advantage is bought by truncating what it is compared against. Companion study, under double-anonymous review at the time of writing, so author and venue are withheld here rather than in the interest of our own anonymity: naming them would compromise a submission that is not this one. The two numbers this paper reuses — the power calibration and the tool-exposure asymmetry — do not depend on that manuscript being available: both recompute from the snapshot of its cells included in this paper’s replication package, with their per-turn logs, 2026.
- Sebastian Baltes, Florian Angermeir, Chetan Arora, Marvin Muñoz Barón, Chunyang Chen, Lukas Böhme, Fabio Calefato, Neil Ernst, Davide Falessi, Brian Fitzgerald, Davide Fucci, Junda He, Christoph Treude, Marcos Kalinowski, Stefano Lambiase, Daniel Russo, Mircea Lungu, Cristina Martinez Montes, Lutz Prechelt, Paul Ralph, Rijnard van Tonder, and Stefan Wagner. Guidelines for empirical studies in software engineering involving large language models. *Empirical Software Engineering*, 2026. Accepted manuscript. Community guidelines from 22 researchers: seven study types and eight guidelines separating requirements (must) from recommendations (should).
- Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9263–9274, 2020. doi: 10.18653/v1/2020.emnlp-main.745.
- Lorenzo De Tomasi, Claudio Di Sipio, Antiniscia Di Marco, and Phuong T. Nguyen. Simplicity by obfuscation: Evaluating LLM-driven code transformation with semantic elasticity. In *Proc. 29th Int. Conf. on Evaluation and Assessment in Software Engineering (EASE)*, 2025. doi: 10.1145/3756681.3757041.
- Tore Dybå, Vigdis By Kampenes, and Dag I. K. Sjøberg. A systematic review of statistical power in software engineering experiments. *Information and Software Technology*, 48(8): 745–755, 2006. doi: 10.1016/j.infsof.2005.08.009.
- Neil A. Ernst and Maria Teresa Baldassarre. Registered reports in software engineering. *Empirical Software Engineering*, 2023. doi: 10.1007/s10664-022-10277-5.
- Irena Gao, Percy Liang, and Carlos Guestrin. Model equality testing: Which model is this API serving? In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*, 2025.

- Andy Georges, Dries Buytaert, and Lieven Eeckhout. Statistically rigorous Java performance evaluation. In *Proceedings of the 22nd ACM SIGPLAN Conference on Object-Oriented Programming Systems, Languages and Applications (OOPSLA)*, pages 57–76, 2007. doi: 10.1145/1297027.1297033.
- Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference, September 2025. URL <https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/>. Blog post, 10 September 2025.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- Alexandru Iosup, Nezih Yigitbasi, and Dick H. J. Epema. On the performance variability of production cloud services. In *11th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid)*, pages 104–113, 2011. doi: 10.1109/CCGrid.2011.22.
- Nan Jiang, Chengxiao Wang, Kevin Liu, Xiangzhe Xu, Lin Tan, Xiangyu Zhang, and Petr Babkin. Nova: Generative language models for assembly code with hierarchical attention and contrastive learning. In *International Conference on Learning Representations (ICLR)*, 2025.
- Fei Lei, Yibo Yang, Wenxiu Sun, and Dahua Lin. MCPVerse: An expansive, real-world benchmark for agentic tool use, 2025.
- Haorui Li, Zhenghui He, Xuanzi Liu, Yang Xu, Dongsheng Liu, Jiakang Ma, Lupan Wu, Yangjie Wu, Xiongchao Tang, and Tianhui Shi. When is the same model not the same service? A measurement study of hosted open-weight LLM APIs, 2026.
- Zhibo Liu and Shuai Wang. How far we have come: Testing decompilation correctness of C decompilers. In *Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2020.
- James Miller, John Daly, Murray Wood, Marc Roper, and Andrew Brooks. Statistical power and its subcomponents — missing and misunderstood concepts in empirical software engineering research. *Information and Software Technology*, 39(4):285–295, 1997. doi: 10.1016/S0950-5849(96)01139-1.
- Todd Mytkowicz, Amer Diwan, Matthias Hauswirth, and Peter F. Sweeney. Producing wrong data without doing anything obviously wrong! In *Proceedings of the 14th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 265–276, 2009. doi: 10.1145/1508244.1508275.
- Vishal Pandey and Gopal Singh. FailureAtlas: A taxonomy of failure modes in multi-provider LLM serving infrastructure, 2026.
- David Pape, Jonathan Evertz, and Lea Schönherr. The silent hyperparameter: Quantifying the impact of inference backends on LLM reproducibility, 2026.
- Ishan Patel, Sahil Sen, Elias Lumer, and Vamse Kumar Subbiah. The bitter lesson of tool calling, 2026.
- Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The Berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pages 48371–48392, 2025.

- Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D’Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermiş, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion, 2025.
- Hanzhuo Tan, Qi Luo, Jing Li, and Yuqun Zhang. LLM4Decompile: Decompiling binary code with large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3473–3487, 2024.
- Yilun Yao, Xinyu Tan, Chao-Hsuan Liu, Yaoming Li, Zhengyang Wang, Wenhan Yu, Zhewen Tan, Yuxuan Tian, Guangxiang Zhao, Lin Sun, Xiangzheng Zhang, and Tong Yang. Harness-bench: Measuring harness effects across models in realistic agent workflows, 2026.
- Yunbei Zhang, Janet Wang, Yingqiang Ge, Weijie Xu, Jihun Hamm, and Chandan K. Reddy. Stop comparing LLM agents without disclosing the harness, 2026.